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**An Explainable Multi-Source Spatiotemporal Model for
Traffic Accident Hotspot Prediction and Severity
Classification in Sri Lanka**

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Contents

Chapter 1: Introduction	1
1.1 Background	1
1.2 Research Problem	4
1.3 Objectives of the Project	6
1.4 Research Questions	6
1.5 Scope of the Research	7
1.6 Chapter Summary	8
Chapter 2: Literature Review	9
2.1 Introduction.....	9
2.2 Findings by other researchers.....	10
2.2.1 Machine learning approach to traffic accident analysis and hotspot prediction ..	10
2.2.2 Crash Severity Prediction and Hotspot Identification using Machine Learning and Spatial Analysis.....	11
2.2.3 Machine learning approach for classifying road accident hotspots	12
2.2.4 Sri Lankan Accident Hotspot and Severity Prediction Model.....	13
2.2.5 Explainable AI traffic accident analysis and hotspot prediction	14
2.2.6 Spatiotemporal Risk Mapping of Statewide Weather-related Traffic Crashes: A Machine Learning Approach.....	15
2.3 The Research Gap.....	16
2.4 Summary of the Findings	18
2.5 Chapter Summary	Error! Bookmark not defined.
References.....	22

Table of Figures

Figure 1 Architecture Diagram	Error! Bookmark not defined.
Figure 2 Gantt Chart -Work Plan.....	Error! Bookmark not defined.
Figure 3Progress Monitoring Sheet	Error! Bookmark not defined.

Chapter 1: Introduction

1.1 Background

Each year road traffic accidents cause significant damage to human lives and economy. Accordingly, World Health Organization reports that around 1.19 million loses their life as a result of road traffic accident while causing significant number of injuries to individuals. and damages to properties (World Health Organization, 2023). Not only that, but such accidents can also cause economic burden to both the individuals and economy of a relevant country via healthcare expenses, property damage, emergency response cost and etc. (Tandrayen-Ragoobur, 2024).

As a country, Sri Lanka also records a gradual increase in vehicle accidents each year due to factors like urbanization and increased number of vehicle usage. According to the Police Spokesman ASP F.U. Wootler, the number of deaths caused by accidents has increased by 322 compared to 2024, recording 2710 death in the year 2025 (Administrator, 2026). Also similar to other countries, road traffic accident has a significant impact to Sri Lankan economy as well. According to a study by Bhavan (2019), it is found that 1% increase in fatality index can negatively affect the country by declining its economic growth by 0.79%. Additionally, the study by Bhavan (2019) also confirms that road traffic accident contributes to increased health expenditures per capita, placing further restrains on financial resources. This confirms that road traffic accident cause harm not only to individual but also cause damage to the country's macroeconomic stability.

To reduce negative impacts of road traffic accidents, Sri Lankan government has already taken necessary steps such as introduction of new laws, polices, awareness programs and etc. (Ministry of Transport, n.d.). However, even with such measures, the number of road traffic accidents fluctuates each year due to numerous factors like poor road conditions, reckless driving and excessive speeding (Over 670 Fatal Road Accidents Reported so Far This Year - Sri Lanka Police, 2026). Accordingly, it is clear that such traditional reactive measures alone are not effective, when it comes to road traffic accident preventions in modern Sri Lanka.

However, advancements in computing technologies such as data analytics and machine learning have enabled us with developing predictive frameworks to minimize road traffic accidents. Existing studies have already incorporated machine learning technologies to implement data driven predictive strategies to minimize road traffic accidents. But exploration of such data driven frameworks are extremely limited when it comes to Sri Lanka where accident contributing factors significantly differ from those countries.

Additionally, just incorporating machine learning in safety critical domain has its own problems. One major problem is machine learning model itself are black box systems. This lack of transparency makes it difficult for authorities to incorporate such technologies for accident hotspot prediction. For example, when it comes to safety critical domain such as transportation and road traffic accident, relevant authorities need to know why a certain model made a certain prediction about accident hotspots. However, Explainable Artificial Intelligence frameworks such as SHAP can be incorporated to improve model's interpretability and thereby it's transparency (IBM, 2023).

Furthermore, road traffic accidents are cause by combination of factors such as weather data, geographical factors, spatial and temporal events. Among those, existing studies found that weather data play a major role in road traffic accidents. For example, adverse weather conditions such as heavy rainfall, reduce visibility and fog may increase the risk of road traffic accidents (Gorzalanczyk, 2025). It is also found that integrating such contextual weather data can improve a model's prediction accuracy.

In addition to above discusses factors, road traffics accidents are also not uniformly distributed across locations and time periods. They vary depends on those spatial and temporal features (Wondwossen Taddesse Gedamu et al., 2025). For example, certain location may prone to more accidents during certain time periods or during certain seasonal trends. So, it is very important for authorities to understand how road traffic accident evolves across locations and time periods along with generic accident hotspot prediction. This can be achieved via spatial temporal analysis of accident data, which also improves accident hotspot prediction and severity classification

Accordingly, this research proposes developing of a multi-source machine learning framework for accident risk prediction and spatial analysis in Sri Lanka, which will be integrated with Sri Lankan police accident records and contextual weather data. This framework will help decision making process of the relevant authorities by identifying accident hotspot locations, contributing factors for the accident prediction, dynamic variations across accidents hotspots over a period of time and presenting those information via an interactive dashboard.

1.2 Research Problem

As mentioned earlier, it is very clear that road traffic accidents pose a significant global threat to both human lives and economy of a country. Like every other country Sri Lanka also deals with such negative impacts of road traffic accidents each year. Despite traditional measures such as traffic laws, fines, and signal systems taken by Sri Lankan government to minimize road traffic accidents, statistics shows that number of accidents fluctuates each year with a significant increase in number of road accidents in recent years (Department of Census and Statistics, 2024).

However, with the recent advancements in computing technologies such as predictive analysis using machine learning researchers have already explored using such technologies for accident hotspot prediction and analysis and thereby to enable implementation of predictive framework to minimize road traffic accidents. One of the best examples for this is study done by de et al. (2023), where they developed a predictive framework for accident hotspot prediction in Brazilian Highways using multiple data sources such as location, time and weather conditions. But exploration of such kind of machine learning frameworks are extremely limited when it comes to Sri Lankan context. Moreover, frameworks developed for countries like Brazil are not validated against Sri Lankan context where accident contribution factors such as weather, peak hours and traffic conditions significant differ from such countries.

When it specifically comes to Sri Lankan context, only a limited number of studies have been done on this exact domain. Accordingly, Jayamanna and Kalansooriya (2024) have explored on implementing a machine learning based system that can predict accident hotspots using historical police accident records. But that study was only limited to police accident data source and given no focus when it comes to integrating contextual data sources such as weather data where such contextual data can improve the model's prediction performance significantly (Yan & Shen, 2022). Accordingly, there is a need for developing a machine learning framework with multiple data sources for Sri Lankan context.

Also as it is mentioned earlier, occurrence of road accidents are not uniformly distributed across locations and time periods and they vary depends on the location (spatial) and temporal (time) features. So, understating such variation can be very helpful for authorities when it comes to decision making and implementing preventative strategies related to road traffic accidents. Additionally, analysis of such spatial and temporal variation can contribute greatly to the model's performance (Yan & Shen, 2022). However, to best of my knowledge there are no existing studies have been done on this spatial and temporal analysis when it comes to Sri Lankan context.

As mentioned earlier, another major problem is lack of transparency among machine learning models. As a result safety critical domains such as transportation hesitate to incorporate machine learning predictive frameworks for accident hotspot prediction and severity classification. However, such issues can be addressed by integrating Explainable AI technologies to machine learning framework as these technologies helps to understand contribution of features to a certain prediction (Al-Omari et al., 2025). But, similar to spatial and temporal analysis of road traffic accident-related data, only a limited number of evidence was found regarding incorporating Explainable AI technologies for accident hotspot prediction and severity classification in Sri Lankan context.

Accordingly, Sri Lanka needs a machine learning framework that uses multiple data sources such as police accident data and weather data for accident hotspot prediction and severity classification. Additionally incorporating analysis of spatial and temporal variation in accident related data Explainable AI technologies can improve both model's prediction accuracy and transparency.

1.3 Objectives of the Project

- To collect, preprocess and integrate police accident records with contextual weather data into a centralized data warehouse via an ETL pipeline.
- To capture dynamic accident risk variations across regions and time by constructing spatiotemporal features.
- To develop and compare performance of multiple machine learning models such as Logistic Regression, Random Forest and XGBoost to identify the most suitable model when using multiple data sources to predict accident hotspots and classify severity in Sri Lanka.
- To analyze the contribution of spatial and temporal features such as location, time and weather conditions to the accident hotspot prediction and severity prediction using XAI techniques such as SHAP
- To develop a prototype dashboard with necessary API for visualization of predicted hotspot location along with location based filtering and meta data insights that support decision making.

1.4 Research Questions

- How to collect, preprocess and integrate police accident records with contextual weather data and store in a data warehouse via an ETL pipeline?
- How to capture accident risk variation across regions and time using spatiotemporal features such as time patterns and location based grids?
- What is the best machine learning model among Random Forest, Logistic Regression, XGBoost, when it comes to accident hotspot prediction and severity classification in the Sri Lanka using multiple data sources?
- To what extent do the spatial and temporal features such as location, time and weather conditions, contribute to the prediction performance of the model?
- How can a dashboard and an API be developed to visualize the predicted hotspot location along with location-based filtering and metadata insights to support decision making?

1.5 Scope of the Research

The main objective of the research is to develop a multi source explainable machine learning framework for accident hotspot prediction and severity classification in Sri Lanka along with analysis of dynamic variations across accident hotspots using spatial and temporal analysis of road accident data. For the implementation, historical police accident records along with weather data obtained through external APIs based on accident location, time and data will be used. Also the scope of this research will be limited to a selected set of regions due to data scarcity in Sri Lankan context.

The inputs for this research include accident related data such as location, date and time, severity, vehicle involvement, casualties, and cause of the accident along with the contextual weather data, which includes features like precipitation, visibility and temperature. Once collected, these data will go through an ETL pipeline where preprocessing techniques like extraction, transformation, feature engineering and finally integrating to a centralized data warehouse happen. After this, these data will be used to train and evaluate machine learning models such as random forest, Logistic Regression and XGBoost to identify the most suitable for Sri Lankan context.

The final output of the research will be a prototype dashboard and an API with only core functionalities needed for visualization of predicted hotspot location along with location based filtering and meta data insights that support decision making.

Additionally, this framework is developing for authorities such as traffic police authorities and road safety analyst who can use the system to identify hotspot prone areas, contributing factors for predicted accidents and thereby to support data driven decision making.

Also, it is very important to note that this research will be limited only to historical data of selected regions as there is no any systematic approach to obtain real time or livestream accident data within Sri Lanka.

1.6 Chapter Summary

In today's world, road traffic accidents are considered as one of the major global threats to both human lives and economy of a country. As a country Sri Lanka also experiences negative effects of road traffic accidents due to several factors such as urbanization, increased vehicle usage and etc. Despite the traditional measures such as implementation of laws and regulations, traffic light systems, awareness programs, fine taken by Sri Lankan government, the number of road traffic accidents grow each year.

However, this chapter introduces our research which uses computing technologies such as machine learning and data analytics to address issues of traditional measures taken to minimize negative effects of road traffic accidents. Additionally, it also explores the usage of Explainable AI technologies along with machine learning models to improve their transparency and thereby to reduce difficulties in adapting such technologies in safety critical domain such as transportation and road traffic accidents.

Furthermore, this chapter also highlights, how such implementation are minimally adapted when it comes to Sri Lankan context along with an overview of limitations of existing studies.

In conclusion this chapter introduces the proposed explainable multi-source spatiotemporal model for traffic accident hotspot prediction and severity classification in Sri Lanka along with its objectives, research questions and scope of the research. The next chapter will explore existing literatures related to accident hotspots prediction, machine learning, techniques used in hotspot predictions, spatiotemporal modelling and explainable AI technologies along with identification of research gaps on existing studies in related to our proposed implementation.

Chapter 2: Literature Review

2.1 Introduction

As it is mentioned earlier, road traffic accidents pose a significant threat to human lives and economy of a country. However, researcher have found that it is possible to use computing technologies like machine learning to minimize road traffic accident effectively and efficiently. By using such technologies, it is possible to implement system that can perform tasks like accident hotspot prediction and such implementations can greatly contribute to decision making by authorities, enabling them to take both preventative and effective actions to minimize road traffic accidents.

Accordingly, this chapter will analyze existing studies related to accident hotspot prediction and severity classification using technologies like machine learning, multi-source data integration, Explainable AI, Spatial and Temporal Analysis in related to both global and Sri Lankan context. Furthermore, a comprehensive explanation about research gaps that is found via exploring current studies will also be included in this chapter.

2.2 Findings by other researchers

2.2.1 Machine learning approach to traffic accident analysis and hotspot prediction (Santos et al., 2021)

When it comes to traffic accident analysis and hotspot prediction Santos et al. (2021) has explored using combination of both supervised and unsupervised machine learning technologies for road accident analysis and hotspot prediction in Setúbal district of Portugal.

Accordingly, when it comes to supervised models, Santos et al. (2021) have used several classification technologies such as Decision Trees, Random Forest, Logistic Regression, and Naïve Bayes. Among them, it is found that Random Forest performs better when it comes to accident hotspot prediction. In addition to these classification algorithms, Santos et al. (2021) have also used unsupervised technologies such as Hierarchical Clustering and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) to identify accident prone location clusters. Accordingly, it found that DBSCAN works effectively when it comes to identifying accident prone location clusters.

Furthermore, comparison of multiple machine learning models can be considered as a major strength as it enables evaluation of their performance in accident hotspot prediction. Moreover, it shows that ensemble models such as Random Forest performs well, when it comes to road traffic accident hotspot prediction, enabling us to choose it as a candidate model for our research.

However, this study was conducted in Portugal, which may limit its adaptability when it comes to Sri Lankan context. Also, the study has given limited focus when it comes to Explainable AI techniques and Spatiotemporal modelling of road accident-related traffic data, leaving gaps in implementing such features.

2.2.2 Crash Severity Prediction and Hotspot Identification using Machine Learning and Spatial Analysis (Islam et al., 2022)

Islam et al. (2022) have also explored on traffic accident hotspot and severity prediction using supervised machine learning approaches. Accordingly, this research was based on Al-Ahsa region of Saudi Arabia. Also, in addition to machine learning approached, researchers have also incorporated Geographic Information System (GIS) based spatial analysis.

Researchers have trained and evaluated 3 supervised machine learning approaches for severity prediction, namely Random Forest, XGBoost, and Logistic Regression. Among them, it is found that Random Forest performed best in their context, achieving 94% accuracy. This study again demonstrates efficiency of supervised machine learning approaches for hotspot and severity prediction. Furthermore, in addition to severity prediction, this study has incorporated Geographic Information System (GIS) based spatial analysis techniques such as Global Moran's I and Getis-Ord G^*i for hotspot identification. Accordingly, results of such analysis highlights that the occurrence of road traffic accidents are based on specific locations with higher risk.

Accordingly, this research shows the importance of combining machine learning models with spatial techniques for severity prediction and hotspot identification and how such framework can benefit transportation authorities in decision making and redcuing road traffic accidents.

However, this research is also limited Saudi Arabian context and not validates against Sri Lankan context. Moreover, usage of Explainable AI techniques and analysis of temporal patterns are highly under explored, leaving gap in combining spatiotemporal analysis along with Explainable AI for accident hotspot prediction and severity classification (Islam et al., 2022).

2.2.3 Machine learning approach for classifying road accident hotspots (de et al., 2023)

de et al. (2023) have already conducted research for identifying road traffic accident locations using supervised machine learning approaches. Additionally, they have also used contextual variables such as weather data that may contribute to traffic accidents.

When it comes to supervised machine learning approaches, de et al. (2023) have used Support Vector Machine (SVM), Random Forest, XGBoost, and Artificial Neural Networks (ANNs) and it is found that ANN performed best with a balanced dataset. Overall researcher highlights that supervised machine learning approaches are effective when it comes to identification of accident hotspots using a balance dataset.

Another strength of this research is the incorporation of external variables such as weather data. Accordingly, it is found that incorporating such variables can improve model's prediction performance and this proves that same concept can be applied when it comes to Sri Lankan context as such approaches remain underexplored.

Additionally, this study was also given limited focus when it comes to integrating Explainable AI techniques and implementation of frameworks like this are underexplored in Sri Lankan context where weather patterns and peak hours significantly differ from those countries, leaving a gap in adapting such technologies in Sri Lankan context (de et al., 2023).

2.2.4 Sri Lankan Accident Hotspot and Severity Prediction Model (J.M.C.T. Jayamanna & Kalansooriya, 2024)

As it is mentioned earlier only a limited number of studies have been conducted on traffic accident hotspot prediction. One of such studies are a study done by J.M.C.T. Jayamanna & P. Kalansooriya (2024), where they developed a machine learning based mobile application for accident hotspot prediction and severity classification using supervise machine learning approaches.

Accordingly, regression supervised machine learning approaches such as Random Forest Regression, Support Vector Regression (SVR), Gradient Boosting Regression, Decision Tree Regression, and XGBoost Regression were used and evaluated. Among them based on metrics such as Mean Squared Error (MSE) and Mean Absolute Error (MAE), XGBoost Regression was identified as the model with the best performance and chosen as the candidate model for accident hotspot prediction.

When it comes to severity classification, approaches such as Random Forest, XGBoost, and Multi-Layer Perceptron (MLP) classifiers have been used and evaluated using metrics such as accuracy, precision, recall and F1-score. Accordingly, it is found that Random Forest classifier perform best when it comes to accident severity prediction with an accuracy of 68%.

However, this research was highly limited to a single source historical dataset which was obtained from Mount Lavinia police station. Additionally, this study has given a limited focus when it comes to integrating contextual data sources such as weather data which can improve model's prediction performance. Furthermore, Explainable AI techniques and spatiotemporal analysis of accident data were severely underexplored, leaving a gap in developing multi source framework with Explainable AI technologies such as SHAP and spatiotemporal analysis for accident hotspot prediction and severity classification in Sri Lanka (J.M.C.T. Jayamanna & P. Kalansooriya, 2024).

2.2.5 Explainable AI traffic accident analysis and hotspot prediction (Al Omari et al., 2025)

Even though machine learning models can perform predictions related to certain task, it fails to explain how the prediction is made. This is because machine learning models act as black box systems, limiting their adaptability in safety critical domains such as transportation and road traffic accidents due to limited transparency and interpretability. However, to mitigate these kinds of issues researcher have explored on integrating Explainable AI techniques with road accident hotspot prediction systems.

Accordingly, Al-Omari et al. (2025) have explored on integrating Explainable AI technique called **SHapley Additive exPlanations (SHAP)** with Random Forest to develop a interpretable machine learning model for road accident severity analysis for Irbid region of Jordan. By using SHAP, it is possible to quantify the contribution of each feature for a prediction and thereby to understand the prominent features affecting final accident severity prediction. Accordingly, it is found that speed was the feature that contributed mostly to the traffic accident severity analysis among several environmental and traffic accident-related features.

Accordingly, this study shows the potential of integrating Explainable AI techniques such as SHAP for traffic accident hotspot prediction related tasks. As such integrations, improve model transparency and interpretability by explaining why a certain prediction is made, the relevant users of the system can make informed decisions.

However, such frameworks are also not validated against Sri Lankan context where contribution factors may differ from countries like Jordan, leaving a gap in incorporating such techniques for accident hotspot prediction in Sri Lanka.

2.2.6 Spatiotemporal Risk Mapping of Statewide Weather-related Traffic Crashes: A Machine Learning Approach (Ogungbire & Pulugurtha, 2025).

As it is mentioned earlier, road traffic accidents are not solely depending on just accident related factors. In fact, contextual spatial and temporal feature like location, time, season and weather contribute heavily to a road traffic accident. But most of the above mentioned studies has given less priority for such analysis of spatial and temporal features. However, by addressing these kinds of limitations, Ogungbire & Pulugurtha (2025) has explored on integrating spatiotemporal risk mapping along with weather related traffic accident hotspot prediction in Nort Carolina region of USA.

Accordingly, for the machine learning model XGBoost was used because of it's capabilities on capturing complex nonlinear relationships among accident related data variables. Additionally, they have also integrated spatiotemporal analytical techniques such as space-time cube analysis, DTW, and Getis-Ord Gi* to the machine learning model in order to facilitate analysis of road traffic accident variations across both location (spatial) and time (temporal).

When it comes to the findings of the above research, it shows that how incorporating spatiotemporal analysis along with accident hotspot prediction can improve the prediction accuracy of machine learning model as occurrence traffic accidents are not solely based on just traffic accident related data.

However, similar to the above studies, this is also limited to North Carolina region of USA, limiting its adaptability in Sri Lanka where both traffic related data and spatiotemporal variations of traffic accident may differ from such countries. Additionally this research is only limited to weather related crash data, making it less applicable for more generalized traffic conditions and regions. Accordingly, a gap remains in incorporating such techniques for the prediction of road accidents with more generalized causes other than just weather (Ogungbire & Pulugurtha, 2025).

2.3 The Research Gap

According to some of the researches, it is clear that integrating traffic accident data with contextual data such as weather data can significantly improve the prediction performance of the model (Ogungbire & Pulugurtha, 2025; Santos et al., 2021). But when it comes to Sri Lanka, most of the studies rely only on single data source such as historical police accident record. Existing studies related to hotspot analysis in Sri Lanka also suggest that integrating weather related data can have significant impact to hotspot identification and analysis process (Nalaka et al., 2021). Accordingly, Sri Lanka needs a centralized data warehouse integrated with multiple data sources such as police accident records and weather data.

Also when it comes to real world accident occurrence varies across different time periods due to factors like traffic hours, and weather and such factors contributed greatly to the model's performance prediction (Yan & Shen, 2022). Also according to the study by Ogungbire & Pulugurtha (2025), it is found that spatial and temporal dependencies can improve model's prediction accuracy. However, this research is only limited to weather related crash data and only a limited number of research has been done on capturing those spatiotemporal variations using features such as time patterns and location based grids, when it comes to broader accident hotspot prediction. Accordingly, a gap remains in combining spatial and temporal features of accidents to create a model that can perform more context aware predictions.

Additionally, most of the existing systems in Sri Lanka were given lesser focus to feature contribution and model explainability analysis. Even though predicting a traffic accident itself is very important, authorities need to know why the model considers a certain location as high-risk hotspot, to support decision making. But machine learning models act as black box systems, limiting their adaptability in safety critical areas. However, it is possible to apply Explainable AI techniques such as SHAP to identify contribution of each feature to the prediction score and thereby it is possible to improve the model's interpretability (Hamdan & Sipos, 2025; Omar Ibrahim Aboulola, 2024; Islam et al., 2022). But most of

the existing research in Sri Lanka has not explored the contribution of features such as location, weather, accident timing and more to predicted accident using explainable AI techniques such as SHAP. Therefore, a gap remains when it comes to model's interpretability, which is very important for decision making.

Based on above gaps, this study proposes a multi-source explainable spatiotemporal machine learning framework for accident hotspot prediction and severity classification for Sri Lankan context. Also, unlike exiting studies, it is planned to integrate both police accident records and weather data, thus creating a multi source data warehouse. Additionally spatiotemporal analysis via feature engineering to identify variations across accident hotspots and explainable AI techniques such as SHAP will be integrated to improve the model's prediction performance and interpretability.

2.4 Summary of the Findings

Author	Key Finding	Research Gaps
Santos et al. (2021)	• Random Forest performs best. • Use multi-Source Data, • Clustering for hotspot prediction	• No XAI techniques like SHAP • Limited temporal analysis
de et al. (2023)	• ANN Forest performs best among SVM, RF and ANN. • Limited spatiotemporal analysis	• Limited explainability • Limited spatiotemporal analysis
Islam et al. (2022)	• Random Forest performs best for severity prediction. • GIS based hotspot prediction using Global Moran's I and Getis-Ord G^*i	• No XAI techniques like SHAP • Limited explanatory variables • Limited spatiotemporal analysis
J.M.C.T. Jayamanna & P. Kalansooriya (2024)	• XGBoost performs best for hotspot prediction • Random Forest performs best for severity classification	• Single data source • No XAI techniques • Limited features • No spatiotemporal modelling
Al-Omari et al. (2025)	• Random Forest performs best.	• Limited to Irbid region of Jordan and not

	• Spatial and temporal pattern analysis • Feature Importance analysis via SHAP	validated against Sri Lankan context
Ogungbire & Pulugurtha (2025)	• Spatiotemporal analysis for risk detection.	• Only limited to weather related crash data.

2.5 Chapter Summary

This chapter consists of a literature review of existing studies related to traffic accident hotspot prediction and severity classification using machine learning models along with multi source data, explainable AI techniques and spatiotemporal analysis.

Accordingly, it is clear that machine learning approaches such as Random Forest, XGBoost, Artificial Neural Network and Support Vector Machines have a overall good performance when it comes to accident hotspot prediction and severity classification. Among them , it is also found ensemble models such as Random Forest and XGBoost performs better with a overall high prediction accuracy.

Furthermore, it is also found that integration of contextual data such as weather can improve model's prediction performance. In addition to integration of contextual data, it is also revealed that accidents are not uniformly distributed along space (spatial) and time (temporal), instead they vary dynamically depends on space and time, thus requiring a need for spatiotemporal analysis of accident-related features.

Also, as it is mentioned earlier, machine learning models act as black box systems, limiting their adaptation in safety critical domains such as transportation and road traffic accidents where authorities need to know why a certain prediction is made. However, several studies related to this matter also discussed in chapter, explaining how can we use Explainable AI techniques such as SHAP to improve model's transparency and interpretability and thereby to improve the trust between authorities and the prediction.

However, in this chapter, several gaps related to those existing studies were also discussed. Accordingly, it reveals that most of the existing studies in Sri Lanka given lesser focus to techniques such as multi source data integration, spatiotemporal analysis and integration of Explainable techniques such as SHAP. Thus, this study proposes a multi-source explainable spatiotemporal machine learning framework to predict accidental hotspots and classify severity of them in Sri Lankan context.

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